

Yuvraj Singh

+91 8604622005 | yuvraj495051@gmail.com | GitHub | LeetCode | LinkedIn

PROFESSIONAL SUMMARY

Computer Engineering undergraduate with strong foundations in C++, Data Structures and Algorithms, SQL, and Generative AI. Experienced in developing Retrieval-Augmented Generation (RAG) applications and responsive web solutions. Passionate about Software Development, AI, and problem solving while actively seeking Software Development Internship opportunities.

EDUCATION

G. H. Raisoni College of Engineering and Management, Pune	Expected Graduation: 2028
<i>B.Tech in Computer Engineering — CGPA: 8.54/10</i>	<i>SPPU</i>
Higher Secondary Certificate (HSC)	2024
<i>76.33%</i>	
Secondary School Certificate (SSC)	2022
<i>80.80%</i>	

TECHNICAL SKILLS

Programming Languages: C, C++, SQL
Core CS: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems
AI: Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI (Learning)
Developer Tools: Git, GitHub, MySQL, VS Code, Ollama
Data Analytics: Power BI

PROJECTS

Yuvraj AI Engine C++, Ollama, Llama 3.2, RAG	GitHub
- Developed a custom Retrieval-Augmented Generation (RAG) engine by implementing a vector database and similarity search pipeline from scratch in C++. - Integrated a local Llama 3.2 model using Ollama for offline AI-powered question answering without relying on external vector database services. - Implemented vector indexing, nearest-neighbor retrieval, and multiple similarity metrics including Cosine, Euclidean, and Manhattan distance. - Optimized vector retrieval using HNSW indexing and benchmarked performance against brute-force and KD-Tree search algorithms, improving retrieval efficiency.	
Mindura HTML, CSS, JavaScript	GitHub
- Developed the frontend of an AI-assisted cognitive healthcare platform for Smart India Hackathon, enabling early-stage dementia assessment through an intuitive and responsive user interface. - Built responsive dashboards, cognitive assessment pages, caregiver portal, consultation booking, and interactive healthcare UI. - Collaborated in a 6-member team and contributed to frontend development, UI/UX improvements, and basic authentication workflow. - Implemented reusable UI components and optimized responsiveness across desktop and mobile devices.	

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Training & Placement Student Coordinator
- Coordinated placement drives by assisting the Training and Placement Cell in scheduling recruitment activities and communicating updates to students. - Act as a communication bridge between students and the placement cell for placement updates and events.
Former Sports Head
Managed and coordinated college sports events while leading student teams and promoting collaboration, leadership, and event execution.